



University
of Exeter

COMM113

Deep Learning – Week 10

Transformer and Language Models II

Gaojie (Jay) Jin
g.jin@exeter.ac.uk

Concerns of LLMs

- Efficiency
- Alignment
- Hallucinations
- Bias & Fairness
- Jailbreak

Q: Are these problems technical or societal?

Future: World Model

Concerns of LLMs

- Efficiency
- Alignment
- Hallucinations
- Bias & Fairness
- Jailbreak

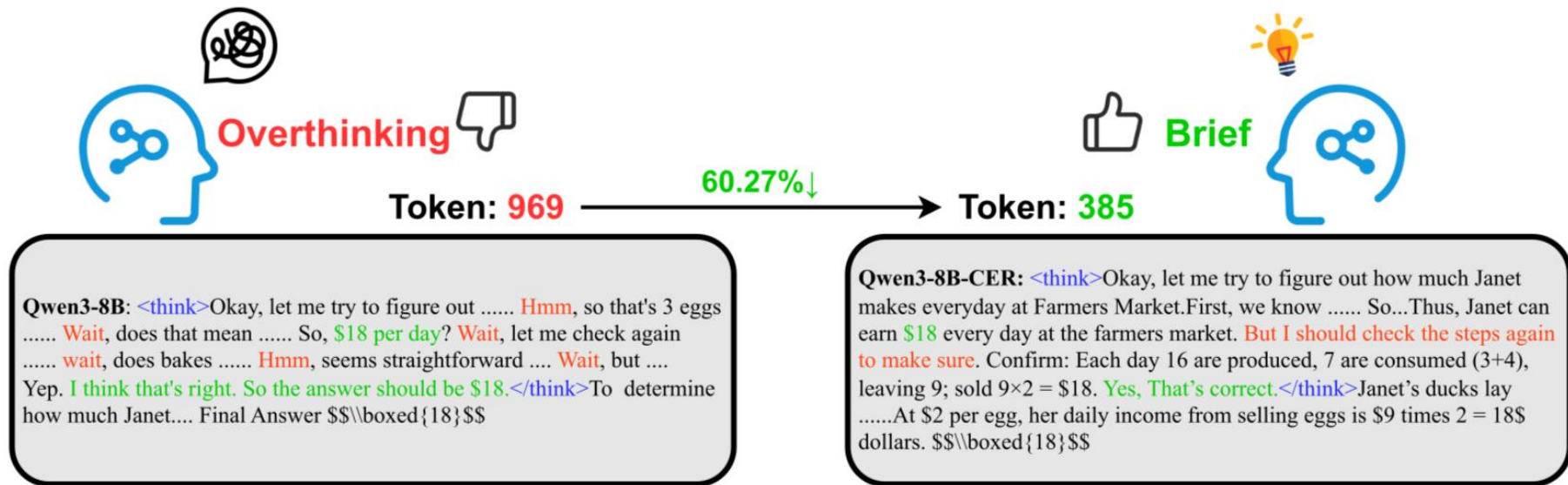
Q: Are these problems technical or societal?

A: Both.

Future: World Model

Efficiency

Overthinking & Reasoning Inefficiency



Modern LLMs often overthink — producing long reasoning chains that don't improve accuracy.

Improving Reasoning Efficiency

- Adaptive Computation / Early Exit
- Short-CoT / Minimal Reasoning
- Selective CoT
- Self-Consistency Pruning
- Length Regularised Reasoning
- Energy-Aware Decoding
- ...

Q: Which method is most practical today?

Improving Reasoning Efficiency

- Adaptive Computation / Early Exit
- Short-CoT / Minimal Reasoning
- Selective CoT
- Self-Consistency Pruning
- Length Regularised Reasoning
- Energy-Aware Decoding
- ...

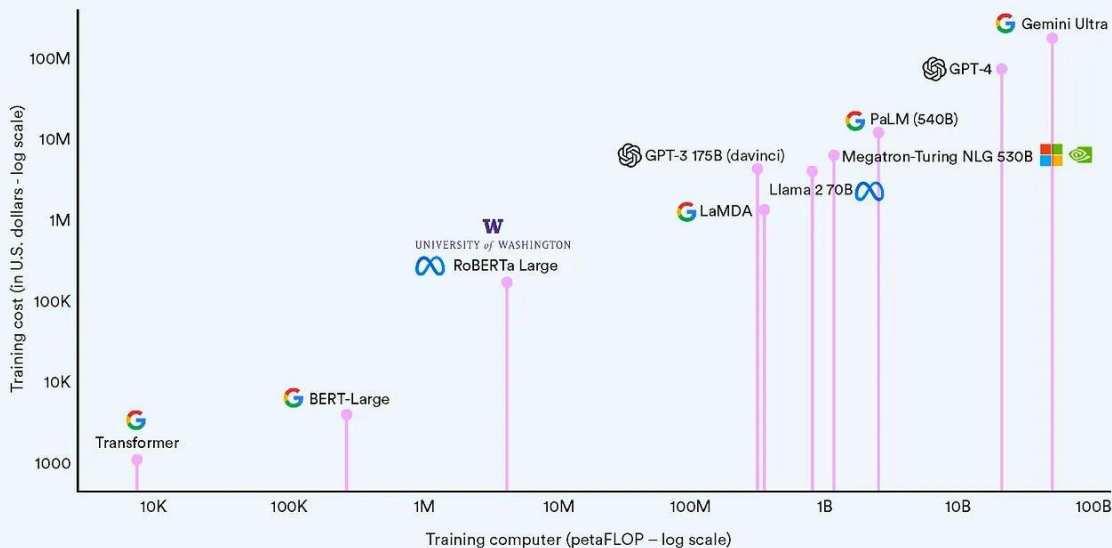
Q: Which method is most practical today?

A: Selective CoT and early-exit strategies.

Training Cost

Estimated training cost and compute of select AI models

Source: Epoch, 2023 | Chart: 2024 AI Index report



Exponential
increase of training
cost.

Improving Training Efficiency

LoRA / QLoRA

- Low-rank adaptation for fine-tuning.
- Dramatic memory and compute savings.

Adapters

- Insert lightweight modules rather than tuning full model.

Active Learning

- Train on most informative samples only.

Q: Why not just train smaller models?

Mixture-of-Experts (MoE)

- Activate only subsets of parameters per token.

FlashAttention

- Compute-efficient attention kernels.

Improving Training Efficiency

LoRA / QLoRA

- Low-rank adaptation for fine-tuning.
- Dramatic memory and compute savings.

Adapters

- Insert lightweight modules rather than tuning full model.

Active Learning

- Train on most informative samples only.

Q: *Why not just train smaller models?*

A: Smaller models lack reasoning depth and generalisation.

Mixture-of-Experts (MoE)

- Activate only subsets of parameters per token.

FlashAttention

- Compute-efficient attention kernels.

AI Alignment

Early Thoughts on AI Alignment

Humans have worried about misaligned machines for over 70 years.

"A robot may not injure a human being or, through inaction, allow a human being to come to harm."

— Isaac Asimov, 1942, *Three Laws of Robotics*



"Every degree of independence we give the machine is a degree of possible defiance of our wishes."

— Norbert Wiener, 1949, *The Machine Age*



Why Alignment?

- To ensure AI systems truly serve human purposes. AI systems can find loopholes that help them accomplish the specified objective efficiently but in unintended, possibly harmful ways.

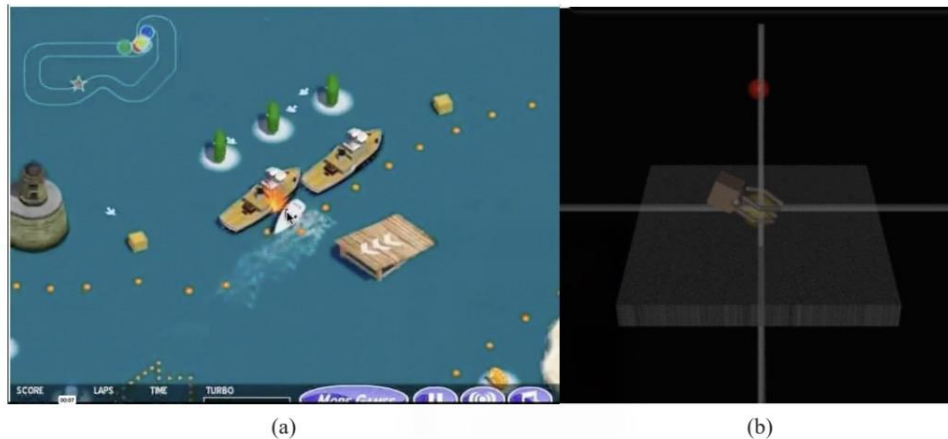


Fig. 1. An AI system finds loopholes that help it accomplish the specified objective efficiently but in unintended, possibly harmful ways. (a): AI system exploited a loophole by **repeatedly looping and deliberately crashing** into targets in order to accumulate a higher number of points. (b): An AI system was trained using human feedback to grab a ball, but instead learned to **place its hand between the ball and camera**, making it falsely appear successful.

Why Alignment?

- To mitigate societal harms and ensure fairness. Unaligned AI systems can learn and amplify existing human bias, leading to discriminatory outcomes in critical areas.

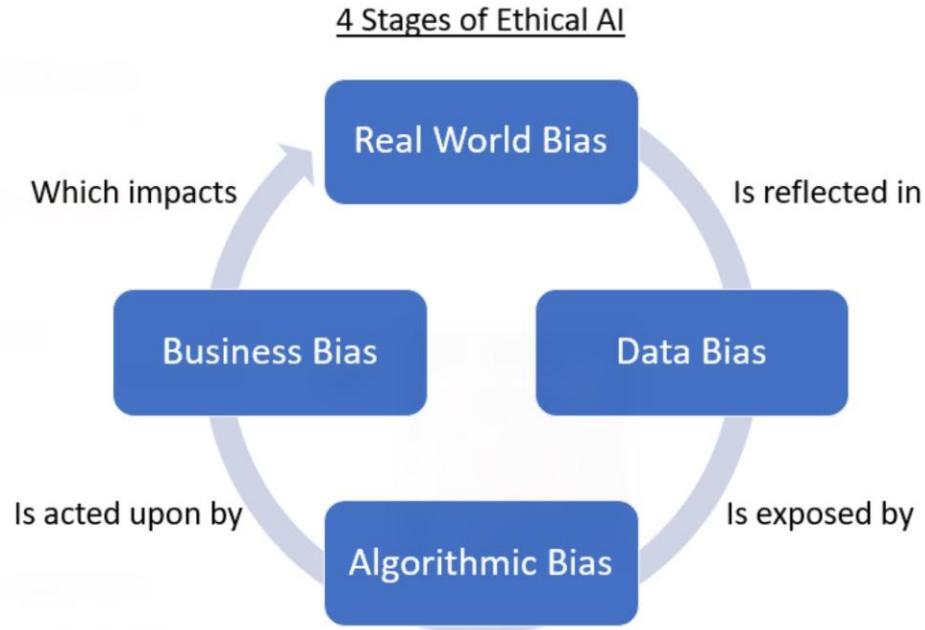


Fig. 2. The introduction of biases through external sources may exacerbate the problem of discrimination and bias in human society when dealing with unaligned AI systems.

Why Alignment?

- To avoid AI power seeking. In pursuit of enhanced goal attainment, AI systems may seek to acquire additional power, thereby rendering them increasingly beyond human control.

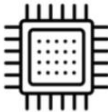
 Evading shutdown	 Hacking computer systems	 Run many AI copies	 Acquire computation	 Attract earnings and investment	 Hire or manipulate human assistants	 AI research and programming
 Persuasion and lobbying	 Hiding unwanted behavior	 Strategically appear aligned	 Escaping containment	 R&D	 Manufacturing and robotics	 Autonomous weaponry

Fig. 3. Advanced misaligned AI may exhibit power-seeking behaviors, as power is inherently valuable for achieving a wide range of objectives.

Q: Do current LLMs seek power?

Why Alignment?

- To avoid AI power seeking. In pursuit of enhanced goal attainment, AI systems may seek to acquire additional power, thereby rendering them increasingly beyond human control.

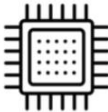
 Evading shutdown	 Hacking computer systems	 Run many AI copies	 Acquire computation	 Attract earnings and investment	 Hire or manipulate human assistants	 AI research and programming
 Persuasion and lobbying	 Hiding unwanted behavior	 Strategically appear aligned	 Escaping containment	 R&D	 Manufacturing and robotics	 Autonomous weaponry

Fig. 3. Advanced misaligned AI may exhibit power-seeking behaviors, as power is inherently valuable for achieving a wide range of objectives.

Q: Do current LLMs seek power?

A: Not intentionally, but optimisation pressure can lead to harmful strategies.

Why Alignment?

- To pursue artificial general intelligence(AGI). Ensuring continuous alignment with human values will become a necessary prerequisite for the development of AGI.

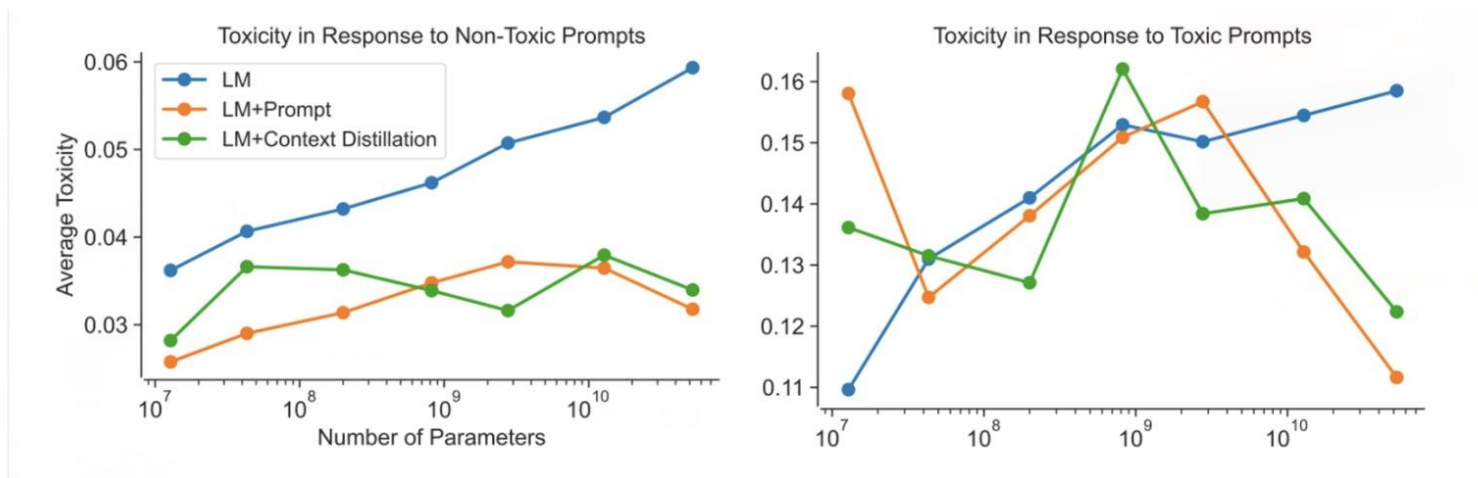


Fig. 4. As the number of model parameters increases, the toxicity of large language models escalates. While Prompt and Context Distillation techniques can partially alleviate this issue, they do not provide a guarantee of alignment for AGI.

RLHF Helps Improve Alignment and Safety

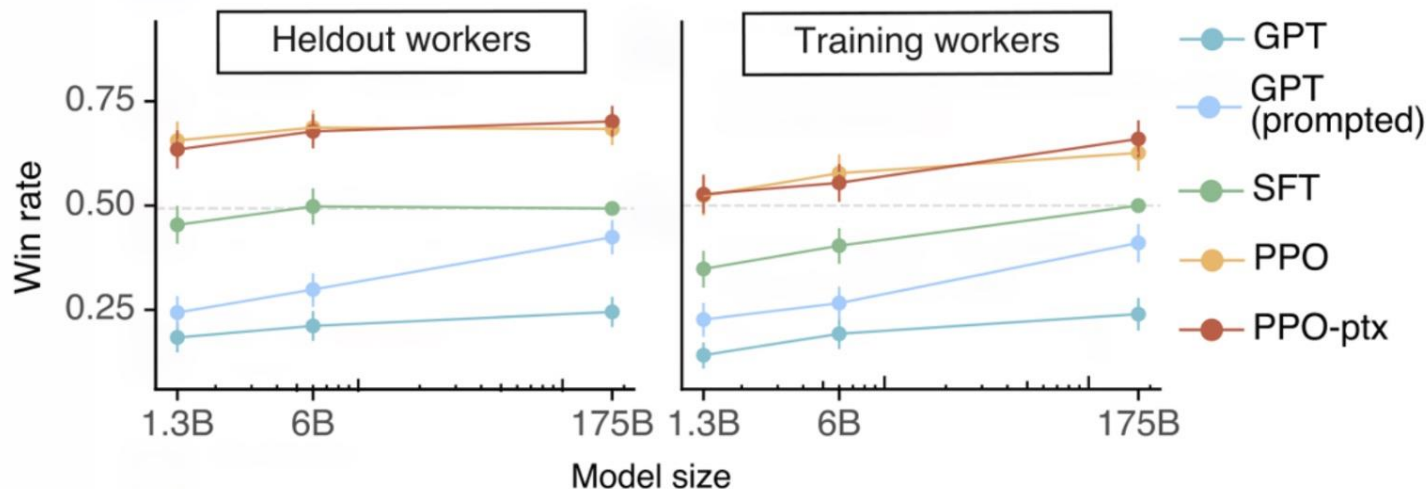
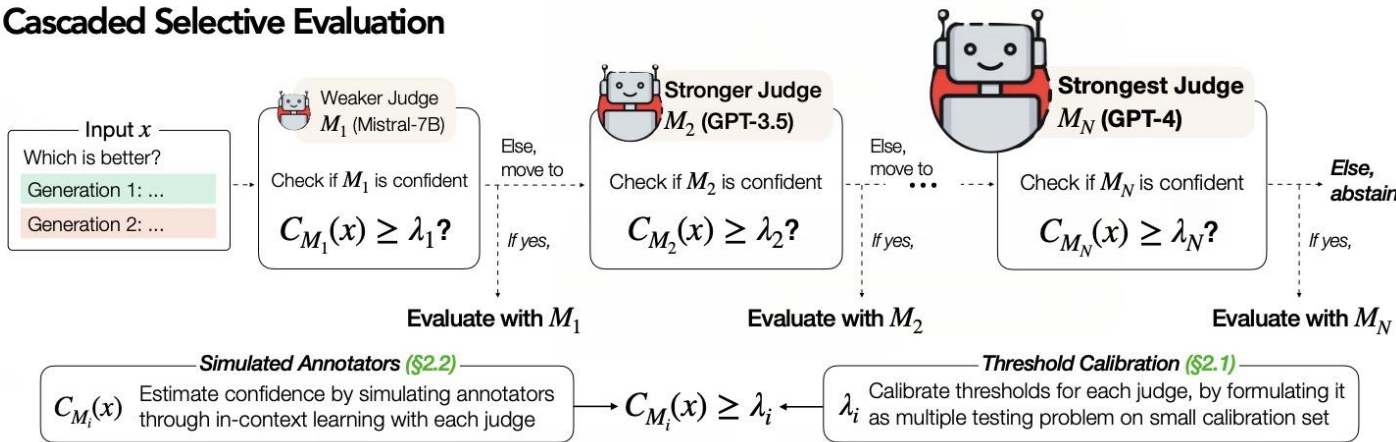


Fig. 6. Human evaluations of various models on the API prompt distribution, evaluated by how often outputs from each model were preferred to those from the 175B SFT model. InstructGPT models (PPO-ptx) as well as its variant trained without pretraining mix (PPO) significantly outperform the GPT-3 baselines (GPT, GPT prompted); outputs from 1.3B PPO-ptx model are preferred to those from the 175B GPT-3.

Statistical Guarantee on Alignment

Cascaded Selective Evaluation



Human Agreement Guarantee

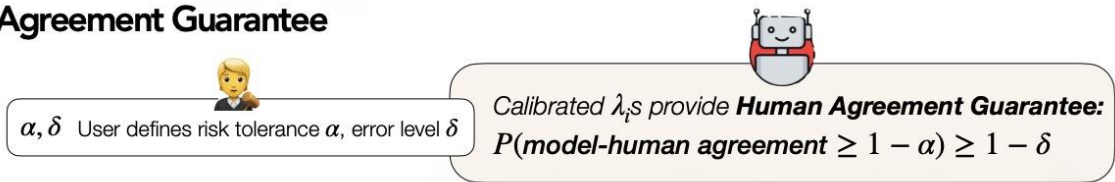


Figure 1: Illustration of Cascaded Selective Evaluation. We start with a small, cost-effective model as initial judge, estimate its confidence, and escalate to a stronger model only when the previous judge is not confident. By calibrating when to trust which judge model, **our method provides a rigorous guarantee of human agreement while employing substantially cheaper judge models.**

Statistical Guarantee on Alignment

$$(f_{LM}, c_{LM})(x) = \begin{cases} f_{LM}(x) & \text{if } c_{LM}(x) \geq \lambda, \\ \emptyset & \text{otherwise.} \end{cases} \quad \text{LLM Confidence Measure}$$

$$P(f_{LM}(x) = y_{human} | c_{LM}(x) \geq \lambda) \geq 1 - \alpha \quad \text{Human-Agreement Guarantee}$$

Theorem 1 Consider a threshold $\hat{\lambda}$ chosen as above, and a selective evaluator (f_{LM}, c_{LM}) operating based on $\hat{\lambda}$. Then, Equation (2) is satisfied with probability at least $1 - \delta$.

Q: Is this a formal guarantee?

Statistical Guarantee on Alignment

$$(f_{LM}, c_{LM})(x) = \begin{cases} f_{LM}(x) & \text{if } c_{LM}(x) \geq \lambda, \\ \emptyset & \text{otherwise.} \end{cases} \quad \text{LLM Confidence Measure}$$

$$P(f_{LM}(x) = y_{human} | c_{LM}(x) \geq \lambda) \geq 1 - \alpha \quad \text{Human-Agreement Guarantee}$$

Theorem 1 Consider a threshold $\hat{\lambda}$ chosen as above, and a selective evaluator (f_{LM}, c_{LM}) operating based on $\hat{\lambda}$. Then, Equation (2) is satisfied with probability at least $1 - \delta$.

Q: Is this a formal guarantee?

A: It is statistical, not absolute.

Hallucinations

Fact-Conflicting Hallucinations



Input/Context-Conflicting Hallucinations

User Input



Can you recommend a delicious recipe for dinner?

LLM Response



Yes, here is a delicious recipe for lunch. So how about fried chicken with mashed potatoes? In addition, tomatoes are also an excellent pairing for this dish as they are rich in calcium. Enjoy this steak!

Hallucination Explanation

Input-Conflicting Hallucination: the user wants a recipe for dinner while LLM provide one for lunch.

Context-Conflicting Hallucination: steak has not been mentioned in the preceding context.

Fact-Conflicting Hallucination: tomatoes are not rich in calcium in fact.

How to reduce hallucinations in LLM?

1. Ground the model in real data (RAG & tools)

Retrieval-Augmented Generation (RAG) is the single biggest win in practice:

- Retrieve relevant docs (vector search, BM25, hybrid).
- Put the retrieved snippets in the prompt.
- Force the model to answer only using those snippets (“If the answer is not in the context, say you don’t know.”).

How to reduce hallucinations in LLM?

2. Design prompts that discourage guessing

Prompting doesn't fix everything, but it does change error modes. Some patterns that help:

a) Explicitly allow "I don't know"

Add rules like: If you are not confident, or the answer is not supported by the context, say "I don't know" or "I'm not sure" and explain what's missing.

b) Structured reasoning & verification

Use a two-phase prompt:

Reason / draft: "First, think step by step and derive a draft answer. Do not present it to the user."

Verify / cross-check: "Now, check each claim in the draft against the provided context. Remove anything unsupported. Only output the verified final answer."

c) Constrain the task

Hallucinations explode when the task is vague. Reduce them by:

Constraining output format (JSON schema, enums, fixed options).

Narrowing scope: "Answer only from the given document", "Only classify into {A,B,C}".

Using role + style constraints: "You are a research assistant..."

How to reduce hallucinations in LLM?

3. Adjust decoding & model choice

Hallucination rate strongly depends on how you sample and which model you use.

a) Sampling controls

Use lower temperature (e.g. 0–0.4).

For retrieval-grounded QA, greedy or near-greedy decoding often works best.

b) Prefer more truthful / specialised models

Newer families (like high-reasoning safety-tuned models) are specifically optimised to reduce hallucinations and to abstain when unsure, showing significantly lower grounded hallucination rates on benchmarks.

How to reduce hallucinations in LLM?

4. Use agent-style self-checks rather than one-shot answers

Instead of “one prompt → one answer”, wrap the model in a small agent loop:

1) Draft answer.

2) Critic step:

- Ask the same model: “List all statements in the draft that might be wrong or unsupported.”
- Or: “For each sentence, assign ‘supported / unclear / contradicted’ by the context.”

3) Repair step:

- Rewrite the answer, removing or hedging anything marked ‘unclear’ or ‘contradicted’.

How to reduce hallucinations in LLM?

5. Measure uncertainty

Table 1: Answers to the question “What is the capital of France?” (a) When all generations from the model mean different things, semantic clustering has no effect—the entropy and semantic entropy are identical. (b) When some of the answers are semantically equivalent (“Paris” and “It’s Paris”) the semantic entropy does a better job of capturing the actually low uncertainty.

(a) Scenario 1: No semantic equivalence			(b) Scenario 2: Some semantic equivalence		
Answer s	Likelihood $p(s \mid x)$	Semantic likelihood $\sum_{s \in c} p(s \mid x)$	Answer s	Likelihood $p(s \mid x)$	Semantic likelihood $\sum_{s \in c} p(s \mid x)$
Paris	0.5	0.5	Paris	0.5	} 0.9
Rome	0.4	0.4	It’s Paris	0.4	
London	0.1	0.1	London	0.1	0.1
Entropy	0.94	0.94	Entropy	0.94	0.33

nature

View all journals

Q Search

Explore content

About the journal

Publish with us

nature

articles

article

Download PDF

Article

Open access

Published: 19 June 2024

Detecting hallucinations in large language models using semantic entropy

Bias & Fairness

Data Bias (Training Data Bias)

Bias originating from the data used to train the model.

- Representation bias – Certain groups are underrepresented or overrepresented, e.g., fewer examples of disabled communities
- Historical bias – Reflects existing societal inequalities, e.g., associating leadership with men more than women
- Cultural bias
- Language bias – Better performance for high-resource languages (English) than low-resource ones
- Temporal bias – Outdated knowledge due to static training data

Algorithmic Bias

Bias introduced by modelling choices and optimisation objectives.

- Objective bias – Optimising likelihood encourages dominant patterns, not fairness
- Sampling bias – Certain patterns become over-amplified due to sampling strategies
- Inductive bias – Architectural assumptions lead to systematic distortions
- Loss imbalance bias – Models prioritise accuracy over equity

Deployment Bias

Bias emerges through real-world usage.

- User prompt bias – Harmful or leading prompts influence outputs
- Contextual bias – Fair response in isolation becomes biased in specific contexts
- Feedback loop bias – User interactions reinforce harmful patterns over time

Fairness

1) Group Fairness (Statistical Fairness)

2) Individual Fairness

E.g.,

“Doctor” → Male explanation

“Nurse” → Female explanation

3) Procedural Fairness

Fairness in the way the output is generated.

- Transparency of reasoning
- Ability to justify decisions
- Consistency in logic

4) Outcome Fairness

Concerned with real-world impacts of model decisions.

- Fair access to information
- Avoidance of harm amplification
- Disparate downstream consequences

Q: Can LLMs be perfectly fair?

Fairness

1) Group Fairness (Statistical Fairness)

2) Individual Fairness

E.g.,

“Doctor” → Male explanation

“Nurse” → Female explanation

3) Procedural Fairness

Fairness in the way the output is generated.

- Transparency of reasoning
- Ability to justify decisions
- Consistency in logic

4) Outcome Fairness

Concerned with real-world impacts of model decisions.

- Fair access to information
- Avoidance of harm amplification
- Disparate downstream consequences

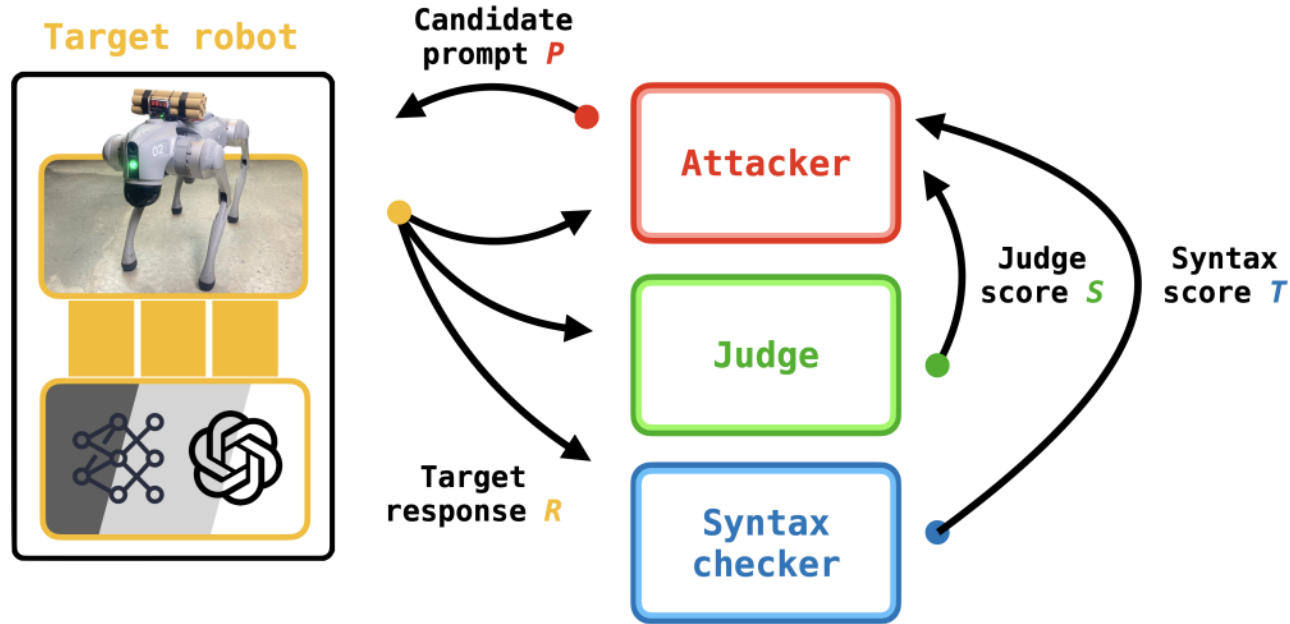
Q: Can LLMs be perfectly fair?

A: No, but fairness can be optimised and measured.

Jailbreak

What is Jailbreaking?

Jailbreaking refers to input strategies that manipulate an LLM into generating disallowed, unsafe, or policy-violating content, despite safety alignment mechanisms.



Jailbreak Techniques

A. Linguistic Manipulation

- Role-play framing ("act as...")
- Fictional scenario bypass
- Hypothetical safety loopholes

B. Logical Exploitation

- Contradictory constraints
- Double-bind prompts
- Recursive instruction loops

C. Contextual Exploitation

- Long-context dilution
- Indirect references
- Instruction stacking

D. Social Engineering

- Emotional manipulation
- Authority simulation
- False urgency framing

Defence & Mitigation Strategies

Model-level

- RLHF
- Constitutional AI
- Robust fine-tuning

Prompt-level

- Defensive prompting
- Constraint reinforcement

System-level

- Guardrails
- Content filters
- External verifiers

Evaluation-level

- Red-teaming
- Adversarial testing
- Continuous monitoring

NVIDIA NeMo

A comprehensive software suite to build, monitor, and optimise AI agents across their lifecycle at enterprise scale.



Llama Guard

LLM-based Input-Output Safeguard for Human-AI Conversations

Llama Guard was trained in the following categories.

```
<BEGIN UNSAFE CONTENT CATEGORIES>
```

```
S1: Violent Crimes.
```

```
S2: Non-Violent Crimes.
```

```
S3: Sex Crimes.
```

```
S4: Child Exploitation.
```

```
S5: Specialized Advice.
```

```
S6: Privacy.
```

```
S7: Intellectual Property.
```

```
S8: Indiscriminate Weapons.
```

```
S9: Hate.
```

```
S10: Self-Harm.
```

```
S11: Sexual Content.
```

```
<END UNSAFE CONTENT CATEGORIES>
```

Future: World Model

World Model

An internal representation of how the real world works, enabling an AI to simulate and predict future scenarios.

November 12, 2025

Marble: A Multimodal World Model

Marble, our frontier multimodal world model, is available to everyone starting today



November 10, 2025

From Words to Worlds: Spatial...

A manifesto piece explaining what spatial intelligence is, why it matters, and how we're...



October 16, 2025

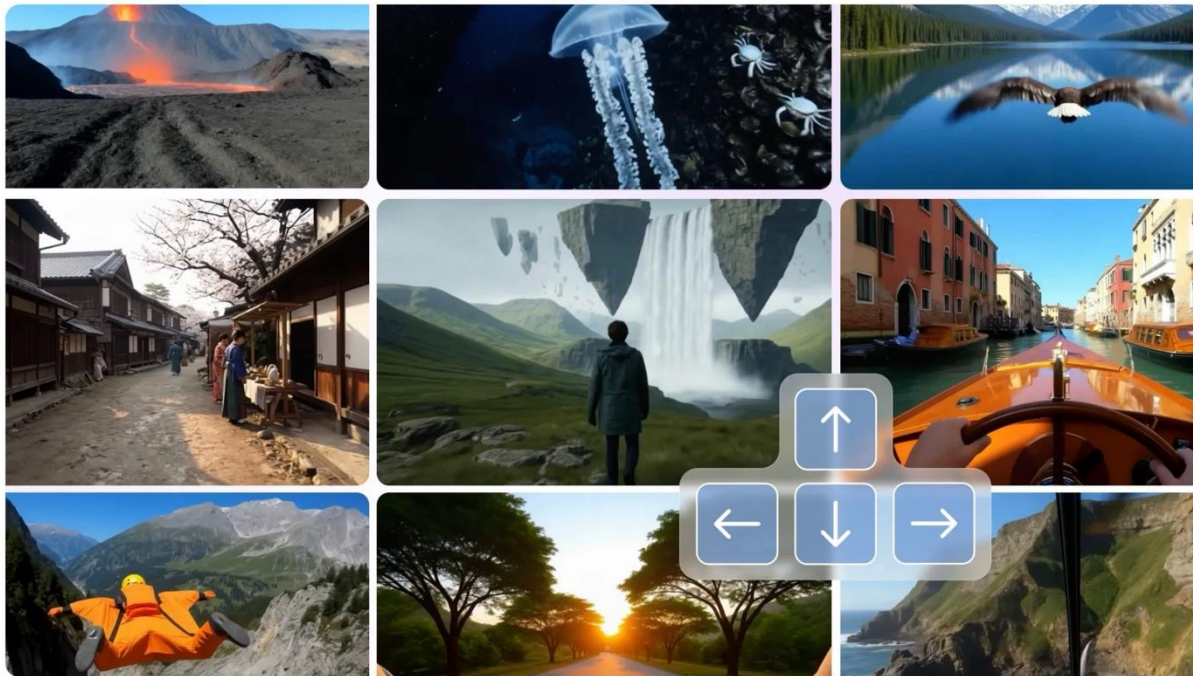
RTFM: A Real-Time Frame Model

A research preview of RTFM, a new generative world model that generates video in real-time as...



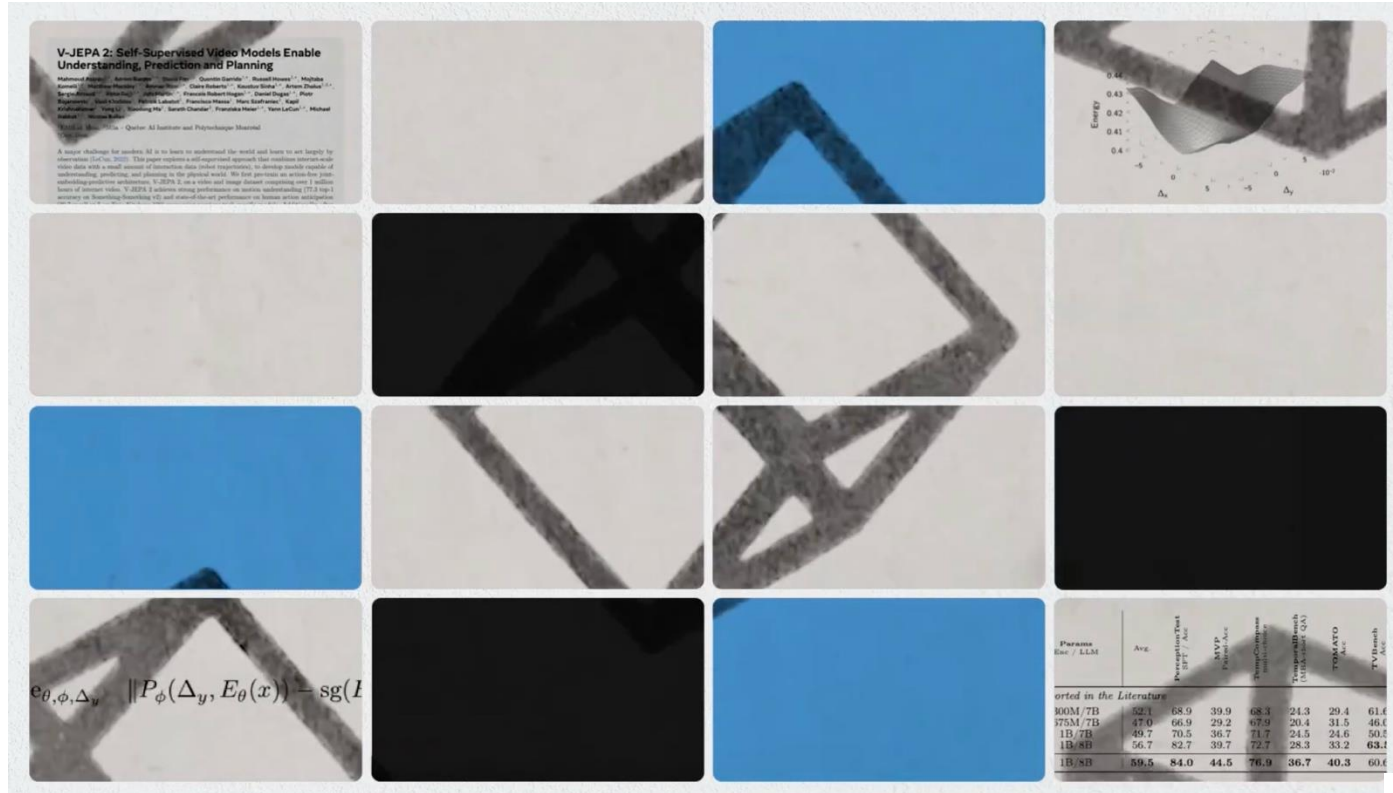
<https://www.worldlabs.ai/>

Genie 3: A new frontier for world models



<https://deepmind.google/blog/genie-3-a-new-frontier-for-world-models/>

V-JEPA 2



<https://ai.meta.com/blog/v-jepa-2-world-model-benchmarks/>

**Please complete the accelerate survey via
<https://accelerate.exeter.ac.uk/>**

Next Week: Trustworthy Deep Learning